

Workshop 1: Hello Quantum World!

QuantUM

The University of Michigan

Fall 2026

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Bits: The Basic Units of Information

Classical computers process classical information, and the fundamental unit of classical information is the **bit**. A single bit can store the answer to one "yes/no" question. We usually represent the two binary states of a bit as "0" and "1".

Binary Numbers

Combining bits enables you to store more information. For example, if you want to store a number from 0 to 15, you could do so with four bits in the following way:

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To convert from a binary number of N bits to a base-10 number, you multiply the least-significant (rightmost) bit by $2^0 = 1$, the next bit to the left by $2^1 = 2$, then the next by $2^2 = 4$, and so on, until you reach the most significant (leftmost bit), which you multiply 2^{N-1} .

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So, that means that N bits can be in one of 2^N different possible states.

Check Your Understanding

How many bits would you need to represent the number 86?

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Solution: Remember, N bits allows you to represent the numbers 0 through $2^N - 1$, so using six bits would get us up to $2^6 - 1 = 63$. That's not quite enough. We add one more bit to get up to $2^7 - 1 = 127$. Now let's break 86 down into powers of 2:

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$$\begin{aligned} 86 &= 64 + 16 + 4 + 2 \\ &= 2^6 \times 1 + 2^5 \times 0 + 2^4 \times 1 + 2^3 \times 0 + 2^2 \times 1 + 2^1 \times 1 + 2^0 \times 0 \\ &= 1010110 \end{aligned}$$

Now, a computer needs to be able to *do* something with the bits in order to, well, compute. Binary gates are the operations that form the fundamental building blocks of all the more complicated algorithms and codes.

Single-Bit Gates: NOT

When you have only one bit, there's only one way you can transform its state: flip the state from 0 to 1 or from 1 to 0. We call this the "NOT" gate. The effect of this gate — and the rest of the gates we'll discuss below — can be represented in a so-called "truth table," with columns for the input and output states of the qubits. The truth table for the NOT gate is:

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input  output

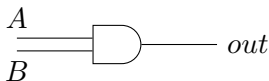
Input	Output
0	1
1	0

Multi-Bit Gates: AND

AND is a two-bit gate that takes two input bits and outputs a single bit. It outputs 1 if both input bits are 1, and 0 otherwise:

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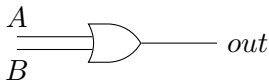
Input A	Input B	Output
0	0	0
0	1	0
1	0	0
1	1	1

Multi-Bit Gates: OR

OR is another two-bit gate with a single output bit. It outputs 1 if *either* of the bits are 1:

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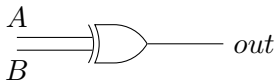
Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

Multi-Bit Gates: XOR

XOR stands for "exclusive OR" and it is like the OR gate, but outputs 1 if *only one* of the input bits are 1. It outputs 0 if they are either both 1 or both 0:

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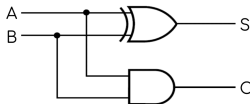
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Input <i>A</i>	Input <i>B</i>	Output
0	0	0
0	1	1
1	0	1
1	1	0

From Gates to Circuits

By combining the gates above, you can do any sort of operation you want on a computer. Let's take a simple example: Using the AND and XOR gates, you can construct the **half-adder circuit**, which calculates the sum of two bits. This is represented in a logical circuit diagram, where the wires represent the bits and the gates operating on the bits are shown as symbols on the corresponding wires:



A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

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What is a Qubit?

Just as bits are the fundamental units of classical information, *quantum* bits, or "qubits," are the fundamental units of quantum information. Like the classical bit, the state of a qubit can be either 0 or 1, which we typically denote as $|0\rangle$ and $|1\rangle$. But unlike the classical bit, a quantum bit can also be in a *superposition* of both the $|0\rangle$ state and the $|1\rangle$ state at the same time.

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In general, a qubit can be in any state $|\psi\rangle$ of the form:

$$|\psi\rangle = c_0|0\rangle + c_1|1\rangle$$

where c_0 and c_1 are complex amplitudes with $|c_0|^2 + |c_1|^2 = 1$.

The Quantum Phase

Because c_0 and c_1 are complex, they can each be written as $c_i = |c_i|e^{i\phi_i}$ where ϕ_i is called the *phase*. If we multiply the whole state by the same overall phase factor, nothing physically changes — this is called a *global phase*, and it has no observable consequences.

For that reason, it is conventional to “factor out” $e^{i\phi_0}$, giving:

$$|\psi\rangle = |c_0||0\rangle + |c_1|e^{i\phi}|1\rangle$$

where $\phi = \phi_1 - \phi_0$ is the *relative* phase of the quantum state, which *does* have observable consequences. This phase plays a very important role in quantum computing, and you will explore its various consequences in the subsequent workshops.

Multiple Qubits

While the state of multiple bits could simply be expressed as a string of 0s and 1s, the state of multiple qubits gets a little more complicated because of the principles of *superposition* and *entanglement*.

Recall that N bits could be in one of 2^N possible states ranging from binary numbers 000...000 to 111...111. But now, because of the principle of superposition, N *qubits* can be in a superposition of *all* of these states at once!

Multiple Qubits: Superposition

The superposition of N qubits can be expressed as

$$\psi_N = \sum_{i=0}^{2^N-1} c_i |i\rangle$$

where, as in the classical case, the state $|i\rangle$ corresponds to the state where each qubit is in the right combination of 0s and 1s to yield the binary number i . These are known as the *computational basis states* of the quantum system. For example, a two-qubit state can be written as a superposition of its four computational basis states:

$$|\psi\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle$$

Each qubit in the system is denoted with an index 0 through $N - 1$. The convention is to read the qubit states from right to left, so that the state of qubit 0 is the rightmost state and the state of qubit $N - 1$ is the leftmost.

Multiple Qubits: Entanglement

As we mentioned earlier, another key feature of qubits is that they can be *entangled* with one another. Let's take an example of a two-qubit state, where $c_0 = c_3 = \frac{1}{\sqrt{2}}$ and $c_1 = c_2 = 0$:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

So, the state of qubit 0 can be either $|0\rangle$ or $|1\rangle$ with equal probability, and same with the state of qubit 1. But these probabilities are no longer independent of one another. If we find the state of qubit 0 to be $|0\rangle$, then we *know* that qubit 1 will also be in $|0\rangle$. This is true no matter how far apart one another, which is why the act of measuring an entangled state is sometimes referred to as "spooky action at a distance."

Multiple Qubits: Entanglement

Entanglement can also take other forms. For example, the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

produces opposite outcomes every time: if one qubit is measured $|0\rangle$, the other is guaranteed to be found in the $|1\rangle$ state.

Check Your Understanding

Is the state $|\psi\rangle = |11\rangle$ entangled? Why or why not?

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In general, if you can describe each qubit's state separately and then multiply them together like this:

$$|\psi\rangle = |\psi_1\rangle|\psi_0\rangle$$

Then it is known as a *product state* and is *not* entangled.

Vector Notation

It is helpful to use vectors and matrices to see how the quantum state transforms under different operations. In this representation, our quantum states will be vectors, and our quantum gates (discussed in the next section) will be matrices that transform the vectors. This is also known as *Dirac notation* or *bra-ket notation*.

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For a single qubit, the vector form of the states are chosen to be:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

In this way, an arbitrary state $|\psi\rangle = a|0\rangle + b|1\rangle$ can be written as

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$$

Vector Notation

For a general, n -qubit state, we will need a 2^n -dimensional vector, with basis states ordered as you might expect, in ascending binary value:

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$$|0 \dots 000\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, |0 \dots 001\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

$$|1 \dots 110\rangle = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \end{pmatrix}, |1 \dots 111\rangle = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Vector Notation

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What are they in ket and vector notation?

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$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \dots, |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

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Quantum gates transform the state of one or more qubits. We represent a gate by a matrix U acting on the state vector:

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A gate acts on every term in a superposition:

$$U(a|0\rangle + b|1\rangle) = aU|0\rangle + bU|1\rangle.$$

Thus, its action on the computational basis determines its action on any state.

Single-Qubit Gates: X

The X gate is the quantum equivalent of the classical NOT gate. It exchanges $|0\rangle$ and $|1\rangle$.

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Input	Output
$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Single-Qubit Gates: Hadamard

The Hadamard gate creates an equal superposition from either computational basis state.

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$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Single-Qubit Gates: Hadamard

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$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Both outputs give 0 or 1 with equal probability when measured in the computational basis. Their *relative phases* differ.

Single-Qubit Gates: Z

The Z gate adds a phase of π to the $|1\rangle$ component, multiplying its amplitude by -1 .

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Input	Output
$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$- 1\rangle$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$Z(a|0\rangle + b|1\rangle) = a|0\rangle - b|1\rangle.$$

The basis measurement probabilities stay the same, but the relative phase can affect later interference.

Single-Qubit Gates: T

The T gate adds a smaller phase shift of $\pi/4$ to the $|1\rangle$ component.

Single-Qubit Gates: T

The T gate adds a smaller phase shift of $\pi/4$ to the $|1\rangle$ component.

Input	Output
$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$e^{i\pi/4} 1\rangle$

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

$$T(a|0\rangle + b|1\rangle) = a|0\rangle + be^{i\pi/4}|1\rangle.$$

Like Z , T changes the relative phase without changing the basis measurement probabilities.

Multi-Qubit Gates: Reversibility

Quantum gates are **reversible**. Their matrices are *unitary*:

$$U^\dagger U = I, \quad U^{-1} = U^\dagger.$$

Here, $U^\dagger$ is the conjugate transpose. Applying the inverse recovers the original state.

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Quantum gates are **reversible**. Their matrices are *unitary*:

$$U^\dagger U = I, \quad U^{-1} = U^\dagger.$$

Here, $U^\dagger$ is the conjugate transpose. Applying the inverse recovers the original state. Classical AND and OR each compress two input bits into one output bit. From that output alone, we cannot reconstruct the inputs. A gate acting on two qubits retains two output qubits. Its matrix has size 4×4 . More generally, an n -qubit gate has a $2^n \times 2^n$ matrix.

Multi-Qubit Gates: CNOT

The controlled-NOT gate flips the **target** if the **control** is $|1\rangle$. The control stays unchanged.

Multi-Qubit Gates: CNOT

The controlled-NOT gate flips the **target** if the **control** is $|1\rangle$. The control stays unchanged. Let $A = q_0$ be the control and $B = q_1$ the target. We write the ket as $|BA\rangle = |q_1q_0\rangle$.

Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 11\rangle$
$ 10\rangle$	$ 10\rangle$
$ 11\rangle$	$ 01\rangle$

$$\text{CNOT}(0, 1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Check Your Understanding: CNOT

Verify by matrix multiplication that $\text{CNOT}(0, 1)$ sends $|01\rangle$ to $|11\rangle$.

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Verify by matrix multiplication that $\text{CNOT}(0, 1)$ sends $|01\rangle$ to $|11\rangle$.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |11\rangle.$$

In $|01\rangle$, the rightmost qubit q_0 is 1, so CNOT flips the leftmost qubit q_1 . **Ordering matters:** a reference that puts the control on the left uses a different matrix for the same controlled operation.

Multi-Qubit Gates: SWAP

The SWAP gate exchanges the states of two qubits.

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Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 10\rangle$
$ 10\rangle$	$ 01\rangle$
$ 11\rangle$	$ 11\rangle$

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Multi-Qubit Gates: Toffoli

The Toffoli gate, or controlled-controlled-NOT (CCNOT), flips the target only when **both controls are 1**.

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The Toffoli gate, or controlled-controlled-NOT (CCNOT), flips the target only when **both controls are 1**. Let $A = q_0$ and $B = q_1$ be controls, and $C = q_2$ the target. Our kets have the order $|CBA\rangle$.

Input	Output
$ 000\rangle$	$ 000\rangle$
$ 001\rangle$	$ 001\rangle$
$ 010\rangle$	$ 010\rangle$
$ 011\rangle$	$ 111\rangle$
$ 100\rangle$	$ 100\rangle$
$ 101\rangle$	$ 101\rangle$
$ 110\rangle$	$ 110\rangle$
$ 111\rangle$	$ 011\rangle$

Toffoli: Matrix Representation

In the ordered basis $|000\rangle, |001\rangle, \dots, |111\rangle$, Toffoli exchanges $|011\rangle$ and $|111\rangle$:

$$\text{CCNOT}(0, 1, 2) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Let's program in Qiskit!

<https://umich-quantum.github.io>